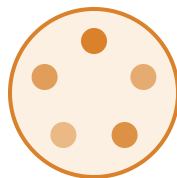


Calibration R^2 is capped by the reference, **not by the model**

THE UNCHECKED PREMISE

One fruit, five destructive determinations – they disagree.



reference value $\bar{y}$
carries its own
measurement noise

Yet R^2 and RMSEP are read as if $\bar{y}$ were exact. In 633 apples the reference's own scatter is **52.30%** of the total.

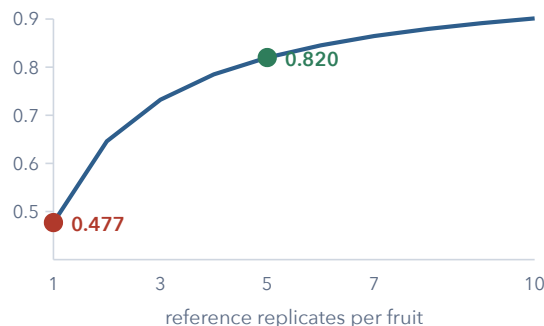


THE ERROR BUDGET

That noise sets a ceiling no model can pass.

0.477 max R^2

against a single-face reference; **0.820** against the five-face mean. Not fitted – it follows from the variance split alone.



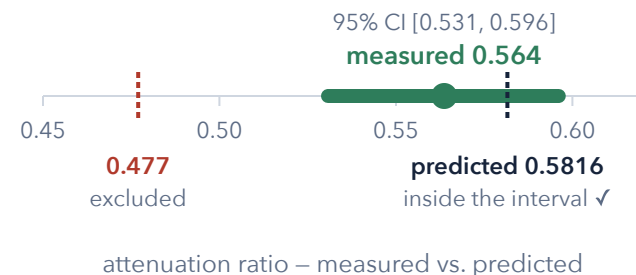
WHAT YOU GET

1 A design rule

Invert the ceiling: name a target R^2 , read off how many replicate determinations your reference needs.

2 A falsifiable diagnostic

Predicts the attenuation ratio with **no free parameter**, so it can be wrong. On 633 apples it was not.



Report performance against a **reference-limited ceiling** – not against unity.